Homework 5

STAT 300 (Fall 2024)

Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirtname.nb.html> file by the deadline.

(1). A company manager is interested in analyzing the relationship between years of working experience and the salary of their employees. He has collected the data from a random sample of employees of their years of experience and the salary. Below provided is partial regression output from R. Use the provided information to answer below questions.

```
##
## Call:
## lm(formula = Salary ~ YearsExperience, data = SalData)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7958.0 -4088.5  -459.9  3372.6 11448.0
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      25792.2     2273.1   11.35 5.51e-12 ***
## YearsExperience   9450.0      378.8   [redacted]
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5788 on 28 degrees of freedom
## Multiple R-squared:  [redacted],  Adjusted R-squared:  0.9554
## F-statistic: [redacted]
```

```
## Analysis of Variance Table
##
## Response: Salary
##                 Df     Sum Sq    Mean Sq F value    Pr(>F)
## YearsExperience  1 2.0857e+10 2.0857e+10 [redacted]
## Residuals       28 9.3813e+08 3.3505e+07
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) Write the equation of the least squares regression line of salary on years of experience.

(b) Interpret the slope and the intercept in the context of the problem.

(c) Calculate the coefficient of determination and interpret this number for this problem?

(d) What is the sample correlation between salary and the years of working experience.

(e) Manger is interested in testing if the salary significantly increases with the years of experience.

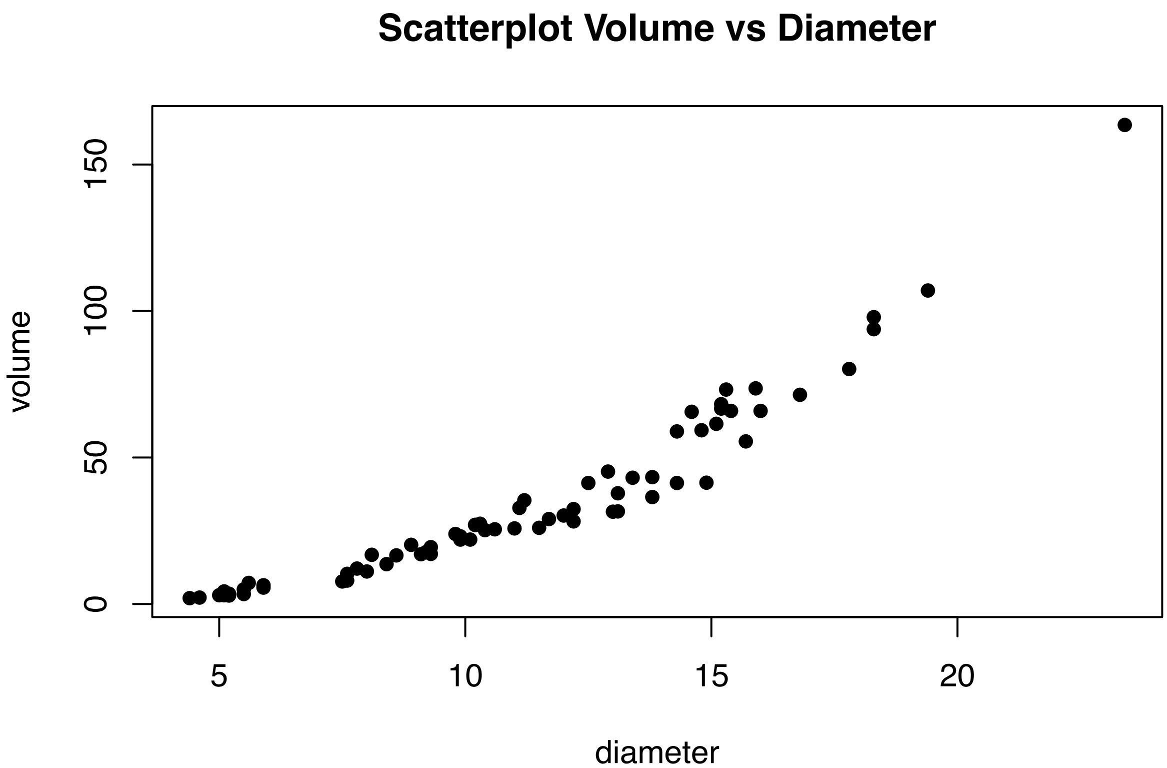
(i) Set up the appropriate H0 and H1 to test this hypotheses.

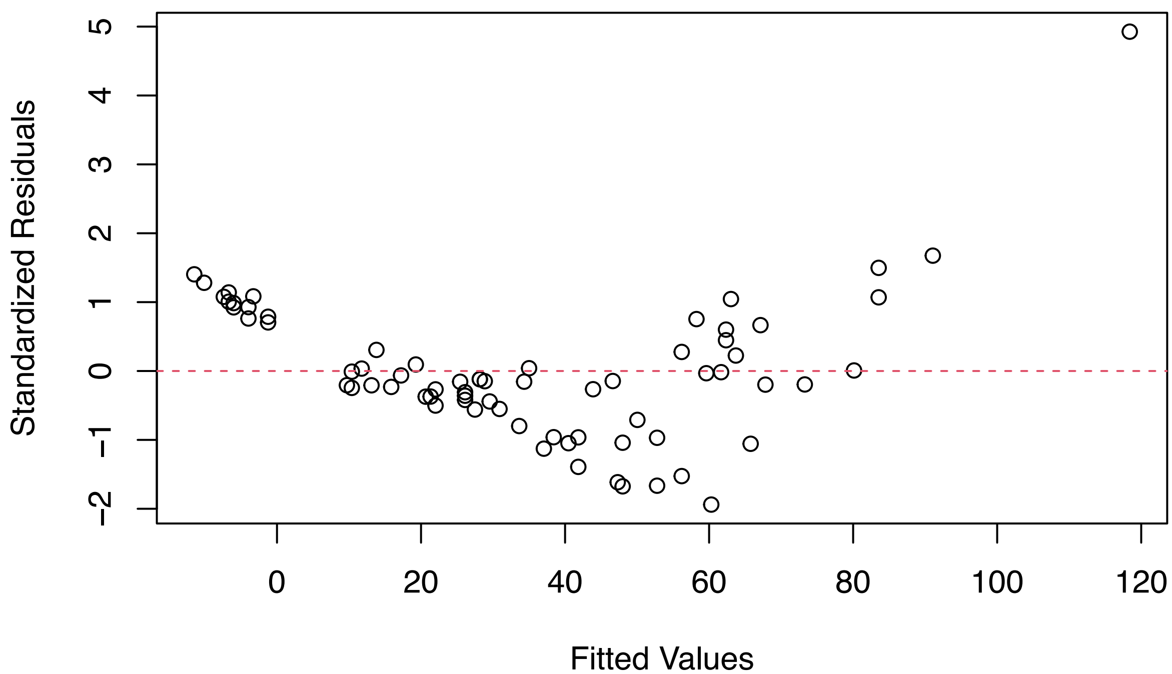
(ii) Calculate the test statistic using the information provided to you in the above partial output.

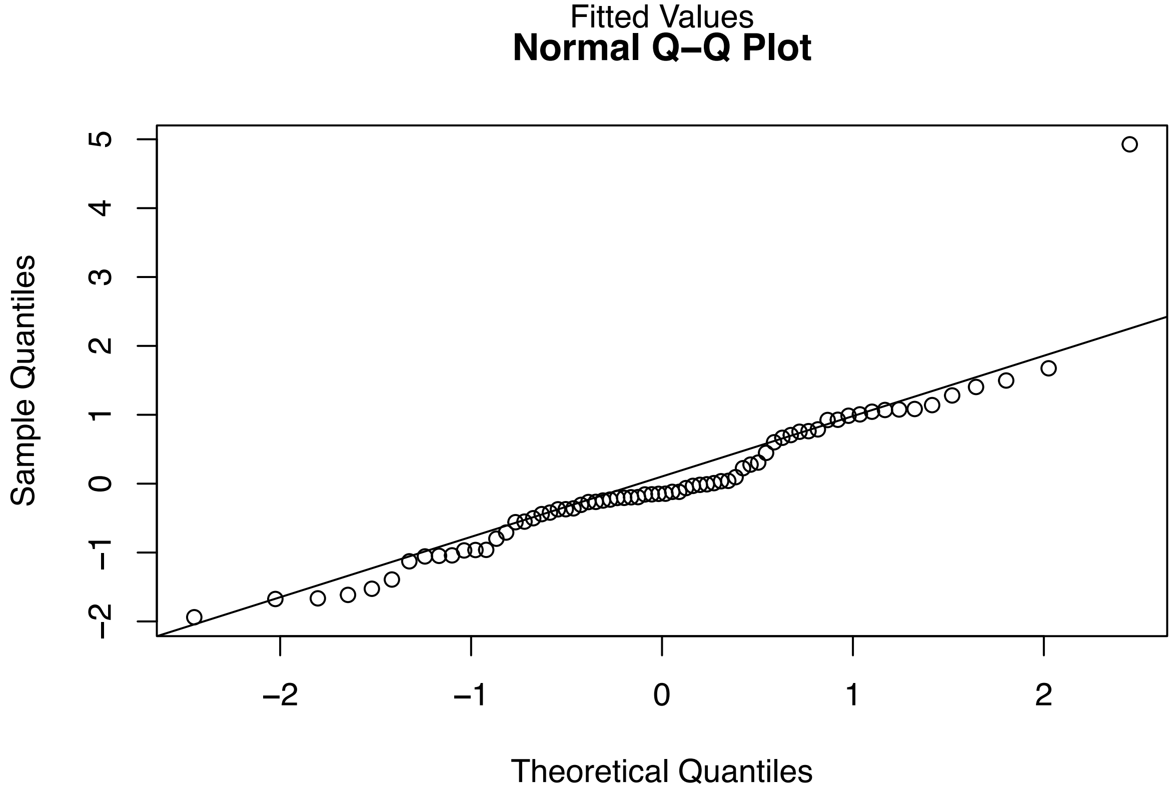
(iii). Calculate the P value for the test. When you write the p value, first write the probability statement corresponding to the p value and then find the numerical value of the p value for the test.

(iv) Write the final conclusion in the context of this problem (use 5 % significance level)

(2). In homework 3 we tried to understand the relationship between the diameter (in inches) and the volume (in cubic feet) of shortleaf pine trees using a random sample of 70 trees. Below provided is a scatterplot, residual plot of the fitted model of volume on diameter and the corresponding QQplot. As you can see, there is clearly a problem with linearity and perhaps even constant variance assumption. Now you will try to improve the model fit and then use it to make inferences.







(a) Make a scatterplot of volume on the log(diameter). Does it looks like it has improved the linearity?

(b) Use log transformation on the response variable as well. Obtain a scatter plot between the two transformed variables. Comment on your observations.

(c) Refit the regression model using the transformed data (i.e.: both transformed) and mark the estimated line on the scatter plot. Comment on your observations. Provide any outputs you might have used.

(d) Check the validity of the assumptions. Use appropriate graphs and comment on each observation.

(e) Test if there is a strong linear relationship between the log(Volume) and the log (Diameter).

  (i) Set up the null and alternative hypotheses for this.

  (ii) From your R output extrat the test statistic for the above test.

  (iii) What is the p value for the test

  (iv) Write the final conclusion in the context of the problem.

(f) Suppose there is a shortleaf pine tree at your home which has a diameter about of about 15 inches. Obtain an interval estimation for the volume of the tree.

(3). The data *storm.csv* file has windspeed (in knots) and the central pressure (in millibars) for a random sample of hurricanes from the past.

(a) Suppose the central pressure is something that can be obtained relatively easily. So the goal is to develop a model to predict the windspeed from the central pressure. Make a scatterplot to display the relationship. Comment on your observations.

(b) Obtain the fitted regression model and overlay it on the scatterplot.

(c) Write the equation of the fitted model.

(d) What percentage of variation in the windspeed is explained by the central pressure of a hurricane?

(e) Test if there is a significant negative correlation (ρ) between the windspeed and the central pressure.

  (i) Set up the correct null and alternative hypotheses.

  (ii) Calculate the test statistic by hand (type up the work). When calculating the test statistic, you mau use R to find the sample correlation, and then show the calculations.

  (iii) What is the distribution of the test statistic under the null hypothesis.

  (iv) What is the p value for the test?

  (v) Write the final conclusion in the context of the problem.

  (vi) Redo the test using R. Include the output and check if your calculations agree with the output.

(f) Obtain a 95% confidence interval for the average windspeed for all the storms with central pressure 920 mb

(g) Obtain a 95% prediction interval for the windspeed for a storm with central pressure 920 mb

(h) Hurricane Katrina had a central pressure of about 920 mb with windspeed 110 knots. Related to the intervals you calculated part (f) and (g), if you were to obtain a rage for the windspeed for hurricane Katrina, which interval you would have used? Explain your reason